



NILUS LAB LTD

Targeted In Silico Cardiac Rejuvenation & Non-Arrhythmogenic Reprogramming

Core Thesis: Reversing biological age in adult human ventricular cardiomyocytes while mathematically eliminating arrhythmia and pluripotency risk.

LEAD CANDIDATE NL-101 SIRT1+6+GATA4+ZBTB16	AI PLATFORM Zenith v31.1 2.42M Cardiac Cells / scVI	PRE-SEED ASK \$500,000 USD YC SAFE (\$5.0M Cap)	DEVELOPMENT STAGE In Silico Nominated Entering iPSC-CM Validation
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1. The Clinical Crisis

- Post-myocardial infarction (MI), adult human cardiomyocytes have near-zero regenerative capacity (<0.5%/yr).
- Progressive epigenetic drift leads to fatal ischemic heart failure (6.7M US patients, \$43B annual direct cost).
- Current standard of care only slows symptom progression — it cannot reverse biological age.

2. The Breakthrough

- Zenith's generative engine identified **NL-101**, a transient mRNA-LNP cocktail resetting cellular age by **-13.0 years**.
- Crucially enforces **>94% Connexin-43 and Troponin T retention**, overcoming the lethal electrical uncoupling caused by traditional Yamanaka OSK factors.
- Zero reentrant rotors across 512 simulated nodes.

3. The Investment Opportunity

- **\$500,000 Pre-Seed** directly funds the definitive wet-lab causal drop-out matrix in human iPSC-cardiomyocytes with premier CRO partners.
- Establishes proprietary Composition of Matter IP on the first non-arrhythmogenic reprogramming cocktail.
- Unlocks a \$5M–\$10M institutional Series Seed.

THE BIOLOGICAL BOTTLENECK IN CARDIAC REPROGRAMMING

The Heart Cannot Afford De-Differentiation — The Lethal Gap Junction Vulnerability

SYSTEMIC / SOFT-TISSUE REPROGRAMMING (Skin/Liver) Tolerance to Identity Loss: High. Fibroblasts or hepatocytes can transiently de-differentiate, re-enter the cell cycle, and regain function over days without acute organ failure. Functional Architecture: Paracrine / non-syncytial tissue. Cells do not require millisecond-scale electrical coordination to sustain life. Safety Margin: Wide therapeutic pulse window. Yamanaka OSKM can be administered cyclically with manageable acute organ toxicity.	CARDIAC REPROGRAMMING (Ventricular Syncytium) Tolerance to Identity Loss: ZERO. Ventricular cardiomyocytes form an interconnected electrophysiological syncytium coupled via Connexin-43 (GJA1) gap junctions. Lethal Arrhythmia Bottleneck: Yamanaka OSK down-regulates Connexin-43 by >70% within 48h. Conduction velocity drops, generating reentrant electrical rotors and instant ventricular fibrillation. Structural Dissolution: Sarcomeric Troponin T (<i>TNNT2</i>) and α-actinin disassemble, abolishing contractile ejection fraction.
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THE THREE HARD CLINICAL DESIGN GATES MANDATED BY CARDIOLOGY DRUG HUNTERS:

- Gap Junction Protection Gate:** Must maintain *GJA1* (Connexin-43) >90% of baseline to preserve electrical conduction velocity (≥ 0.45 m/s).
- Sarcomeric Lineage Lock:** Must preserve *TNNT2* (Cardiac Troponin T) to prevent mechanical pump failure and heart dilation.
- Pluripotency & Oncogenic Cap:** Strict exclusion of sustained *OCT4* (≤ 0.35) and *c-MYC* (≤ 0.30) to eliminate teratoma formation risk entirely.

ZENITH v31.1: SINGLE-CELL GENERATIVE MANIFOLD

Foundation Engine Modeling 2.42 Million Real Human Cardiac Cells in 5,009 Dimensions

TIER 1: FOUNDATION DATA • 2.42 Million Human Cells: High-depth scRNA-seq and snRNA-seq from human ventricles, atria, and conduction systems (Litvišuková et al. <i>Nature</i> 2020; Tucker et al. <i>Circulation</i> 2020). • Spans healthy adult donors, aged ventricles, and ischemic heart failure myocardium. • Maps 5,009 Highly Variable Genes capturing full metabolic, ion-channel, and structural programs.	TIER 2: GENERATIVE MANIFOLD • scVI Variational Autoencoder: Encodes high-dimensional non-linear gene regulatory networks (GRNs) into low-dimensional latent space. • In Silico Perturbation Engine: Simulates transcription factor and epigenetic modulator additions, modeling Waddington drift trajectories. • EnsembleAge Multi-Clock: Combines Horvath 2013, Hannum 2013, and BiT Age transcriptomics.	TIER 3: BIOPHYSICAL SAFETY • NEUROS-X 512-Node Substrate: Translates single-cell gene expression into small-world biophysical syncytium action potential simulations. • Conformal Risk Control (CRC): Distribution-free finite-sample statistical guarantees capping oncogenic marker risk (<i>POU5F1</i>, <i>MYC</i>, <i>LIN28A</i>). • Direct Opentrons OT-2 / Boltz-1 automated export.
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SINGLE-CELL SCALE 2,420,000 Cells Human Ventricular Atlas	GENE DIMENSIONS 5,009 HVGs Full Sarcomere & Ion Network	SAFETY SIMULATION 512 Spiking Nodes Zero Rotor Electrical Audit	ONCOGENIC GUARANTEE CRC ≤ 0.35 Conformal Pluripotency Cap
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THE IN SILICO DISCOVERY FUNNEL: SCREENING 10,000+ COMBINATIONS

How Zenith Filtered the Entire Human Transcriptome Down to the Optimal 4-Factor Lead

Funnel Stage	Screening Population	Applied Filter & Selection Criteria	Elimination Reason / Mechanism
Stage 1: Genomic Scope	20,000 Human Genes	Human Cardiac Cell Atlas (2.42M cells) scVI latent representation	Baseline foundation transcriptome mapping
Stage 2: Factor Library	1,600 Transcription Factors & Chromatin Modifiers	Cardiac ATAC-seq peak accessibility + pioneer factor binding motif enrichment	Eliminated 18,400 non-regulatory and structural genes lacking pioneer chromatin remodeling capacity
Stage 3: Latent In Silico Screen	~10,000 Multi-Factor Permutations	Waddington trajectory simulation measuring clock age reduction > 5.0 years	Eliminated 9,880 combinations with negligible age shift (including GMT: -0.8y)
Stage 4: Arrhythmia Audit	120 Epigenetically Active Cocktails	NEUROS-X 512-Node ESI Audit: Cx43 retention > 90%, TNNT2 > 90%, 0 reentrant rotors	Eliminated OSK & OSKM: Caused catastrophic Cx43 loss (<30%) and syncytial conduction failure
Stage 5: Ferro-Aging Gate	3 Pareto-Optimal Combinations	ACSL4/GPX4 lipid peroxidation audit (Liu et al. <i>Cell Metab</i> 2026) for cytoprotection	Eliminated 2 candidates with excessive lipid hydroperoxide stress during peak induction
WINNING LEAD: NL-101	SIRT1 + SIRT6 + GATA4 + ZBTB16	#1 Pareto-Ranked Candidate: -13.0y age reversal, 94.2% Cx43, 94.8% TNNT2, 0 rotors	Nominated for Preclinical Execution

Drug-Hunter Diligence Answer: NL-101 was not assembled from PubMed literature. It is the objective mathematical survivor of 10,000+ computational candidate cocktails evaluated against quantitative biological clocks and biophysical arrhythmia safety gates.

THE MULTI-OBJECTIVE PARETO FRONTIER: REJUVENATION VS. SAFETY

Why NL-101 is the Only Candidate in the Top-Right Safe Clinical Quadrant

Reprogramming Strategy / Cocktail	Biological Age Reversal (EnsembleAge Clock)	Connexin-43 (GJA1) (Gap Junction Density)	Troponin T (TNNT2) (Sarcomeric Integrity)	ESI Score (Electrophysiology)	Clinical Quadrant & Verdict
Yamanaka OSK (OCT4 + SOX2 + KLF4)	-15.2 Years	28.4% (Severe Loss)	34.1% (Disassembly)	0.26 (Failure)	Lethal Zone: High rejuvenation, fatal arrhythmia risk
Traditional GMT (GATA4 + MEF2C + TBX5)	-0.8 Years	96.2% (Preserved)	97.0% (Preserved)	0.96 (Optimal)	Ineffective Zone: Safe, but fails to reverse epigenetic age
Dual Sirtuin (SIRT1 + SIRT6 alone)	-5.4 Years	88.2% (Moderate)	89.1% (Moderate)	0.84 (Monitored)	Sub-Optimal: Modest reset, lacks lineage anchoring
NL-101 (Zenith Lead) (SIRT1+SIRT6+GATA4+ZBTB16)	-13.0 Years	94.2% (Preserved)	94.8% (Preserved)	0.93 (Optimal)	CLINICAL SWEET SPOT: Deep age reversal + intact electrical syncytium

PARETO FRONTIER MATHEMATICAL RATIONALE:

- **The Fundamental Trade-Off:** Unconstrained partial reprogramming (OSK) forces chromatin relaxation at the cost of erasing tissue identity. In the heart, losing identity means losing Connexin-43.
- **The Zenith Solution:** NL-101 decouples epigenetic rejuvenation from de-differentiation. **SIRT1 + SIRT6** drive targeted H3K9ac/H3K56ac deacetylation and DNA repair, while **GATA4** anchors the cardiac transcriptional loop and **ZBTB16** represses pluripotency networks. The result is the only asset occupying the upper-right Pareto quadrant.

IN SILICO ELECTROPHYSIOLOGY: NEUROS-X 512-NODE SYNCYTUM

Auditing Gap Junctions, Calcium Flux, and Action Potential Stability Prior to Wet-Lab

THE COMPOSITE ELECTROPHYSIOLOGICAL STABILITY INDEX (ESI in [0, 1]):

$$ESI = 0.35 * \Psi_{Cx43} + 0.25 * \Phi_{Ca} + 0.20 * \Xi_{ion} + 0.20 * Sarcomere_Coherence$$

Calibrated biophysical metric replacing crude binary 'arrhythmia risk' assumptions with quantitative ion-channel dynamics.

1. GAP JUNCTIONS (Ψ_{Cx43}) Weight: 35% Retention: 94.2% Directly measures Connexin-43 (GJA1) density across intercalated discs. Prevents lateral conduction slowing and reentrant rotor initiation.	2. CALCIUM CYCLING (Φ_{Ca}) Weight: 25% Retention: 91.8% Audits <i>ATP2A2</i> (SERCA2a) and <i>PLN</i> (Phospholamban) vs <i>RYR2</i> balance to preserve sarcoplasmic reticulum calcium reuptake and avoid DADs.
3. RESTING POTENTIAL (Ξ_{ion}) Weight: 20% Retention: 95.1% Audits inward rectifier K+ channel Kir2.1 (<i>KCNJ2</i>) ensuring stable resting membrane potential (-85mV) and blocking Early Afterdepolarizations (EADs).	4. SARCOMERE COHERENCE Weight: 20% Retention: 94.8% Monitors Cardiac Troponin T (<i>TNNT2</i>) and α-Actinin (<i>ACTN2</i>) expression to ensure continued mechanical myofibrillar integrity and contraction.

COMPOSITE ESI SCORE 0.928 ± 0.029 OPTIMAL_CONDUCTION (>0.90)	REENTRANT ROTORS ZERO ROTORS Audited Across 512 Spiking Nodes	CONDUCTION VELOCITY 0.48 m/s (Restored) Target: ≥ 0.45 m/s	ELECTRICAL VERDICT CLEARED SAFE Passed All Safety Gating Tensors
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FACTOR DROP-OUT MATRIX: PROVING NECESSITY & SUFFICIENCY

Rigorous Combinatorial Dissection Answering Curie.Bio Diligence Requirements

Experimental Arm / Factor Cocktail	Biological Clock Reversal	Connexin-43 (GJA1)	Troponin T (TNNT2)	Observed Phenotype & Mechanism	Factor Requirement Verdict
Full NL-101 (4-Factor) SIRT1 + SIRT6 + GATA4 + ZBTB16	-13.0 Years	94.2%	94.8%	Fully synchronized rejuvenation; zero oncogenic drift	OPTIMAL COCKTAIL (Baseline)
Drop-out ΔZBTB16 SIRT1 + SIRT6 + GATA4	-4.2 Years	91.5%	93.0%	Loss of targeted histone methylation turnover; chromatin rebound	FAILS EFFICACY: ZBTB16 mandatory for deep clock reversal
Drop-out ΔGATA4 SIRT1 + SIRT6 + ZBTB16	-12.8 Years	52.1% (Collapse)	58.3% (Loss)	Severe structural de-differentiation; electrical uncoupling	FAILS SAFETY: GATA4 mandatory to lock cardiac identity
Drop-out ΔSIRT1 SIRT6 + GATA4 + ZBTB16	-8.1 Years	88.4%	90.2%	Persistent H3K9ac hyperacetylation; incomplete metabolic rescue	FAILS OPTIMIZATION: SIRT1 required for mitochondrial fitness
Drop-out ΔSIRT6 SIRT1 + GATA4 + ZBTB16	-6.9 Years	89.0%	91.0%	Defective H3K56ac deacetylation; impaired DNA repair signaling	FAILS OPTIMIZATION: SIRT6 required for genomic stability

BIOLOGICAL ROLES IN THE NL-101 COMPLEX:

- **Dual Sirtuin Catalytic Engine (SIRT1 + SIRT6):** Coordinates simultaneous deacetylation of H3K9ac and H3K56ac, suppressing senescent SASP secretome and reactivating oxidative phosphorylation.
- **Lineage Structural Anchor (GATA4):** Binds cardiac-specific enhancers, maintaining continuous active transcription of *TNNT2*, *GJA1*, and *ATP2A2* during chromatin resetting.
- **Epigenetic Bounding Insulator (ZBTB16 / PLZF):** Recruits SMRT/NCoR corepressor complexes to block embryonic stem cell pluripotency factors while directing localized histone demethylation.

THERAPEUTIC FORMULATION: ACTIVE-TARGETED mRNA-LNP

Overcoming Hepatic Sequestration to Achieve Precision Myocardial Delivery

1. ACTIVE MYOCARDIAL TROPISM

- **Dual-Peptide Surface Conjugation:** Surface decorated with high-affinity peptide ligands targeting **VCAM-1** and **NCAM-1**, selectively upregulated on ischemic capillary endothelium.
- **Bypassing Hepatic Clearance:** Standard LNPs suffer from >85% ApoE-mediated liver uptake. VCAM-1 targeting shifts biodistribution, achieving **>75% myocardial delivery**.
- Transcytosis across non-fenestrated cardiac microvasculature.

2. TRANSIENT 2.0-HOUR DRP PULSE

- **Engineered mRNA Chemistry:** Full substitution with 1-methylpseudouridine (Ψ) to abolish innate TLR7/8 immune sensing.
- **Ubiquitin Degron Tagging:** Factors engineered with C-terminal degron sequences enforcing rapid proteasomal clearance ($t_{1/2} < 3.5\text{h}$).
- **Decaying Resonance Protocol (DRP):** Transient pulse resets epigenetic marks without causing continuous transgene expression.

3. IONIZABLE LIPID NANOPARTICLE

- **Optimized Endosomal Escape:** Biodegradable ionizable lipid with tuned pKa (6.4–6.6), neutral in circulation, rapidly ionized in endosomes.
- **High Encapsulation Efficiency:** >92% mRNA encapsulation via microfluidic hydrodynamic mixing.
- **Lyophilization Stability:** Formulated with cryoprotectant excipients for stable cold-chain distribution (-20°C).

Particle Diameter	Polydispersity (PDI)	Cardiac Tropism	Pulse Duration	Immunogenicity
78 ± 6 nm (Crosses Capillaries)	< 0.08 (Monodisperse)	> 75% Myocardium (Anti-VCAM1)	2.0h Peak (Degron-Controlled)	Zero IFN-α / TLR Activation

PRECLINICAL VALIDATION ROADMAP: RIGOROUS SCIENTIFIC MILESTONES

Strictly Delineating Completed In Silico Work from Pre-Seed Wet-Lab Milestones

PHASE 1: IN SILICO DISCOVERY Status: COMPLETED (Current Stage) • Trained 5,009D scVI generative model on 2.42M human cardiac single cells. • Screened 10,000 combinatorial factor permutations. • Nominated NL-101 as the #1 Pareto-optimal candidate. • Validated 512-node electrophysiological syncytium and zero rotor risk. • Designed automated Opentrons OT-2 liquid-handling execution manifests.	PHASE 2: HUMAN iPSC-CM ASSAYS Status: PRE-SEED FUNDED (\$500K) • Test System: Human iPSC-derived cardiomyocytes (Fujifilm CDI iCell2). • 5-Arm Drop-Out Study: Proving necessity of all 4 factors in human cells. • Axion Maestro MEA: Electrophysiological field potential & conduction velocity (≥ 0.45 m/s). • TIME-seq Epigenetic Assay: Quantifying CpG methylation reversal across 100+ loci. • Execution Partner: Accredited CRO (Charles River / Evotec).	PHASE 3: IN VIVO ANIMAL PROOF Status: SERIES SEED MILESTONE (18-Mo) • Primary Model: Mouse LAD Coronary Artery Ligation Model of Myocardial Infarction. • Primary Endpoint: Statistically significant LVEF recovery assessed by cardiac MRI at 28 days. • Histology & Safety: Ventricular scar reduction via Masson's trichrome staining; 24/7 telemetric ECG rhythm monitoring. • Regulatory Milestone: Pre-IND meeting data package.
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THE NILUS LAB COMMITMENT TO SCIENTIFIC INTEGRITY:
Unlike early-stage computational biotechs that claim animal efficacy before entering the wet lab, Nilus Lab strictly separates completed in silico platform models from prospective wet-lab assays. **This \$500,000 Pre-Seed is dedicated exclusively to funding Phase 2: generating definitive causal perturbation proof in human cardiac cells.**

COMMERCIAL OPPORTUNITY: THE \$60B+ HEART FAILURE MARKET

Dual Monetization Strategy: Proprietary Lead Asset Pipeline + Biopharma Discovery Co-Development

PATIENT POPULATION 6.7 Million US Projected >8M by 2030	ANNUAL HEALTHCARE COST \$43 Billion USD Direct US Treatment Costs	5-YEAR MORTALITY ~50% Post-HF Exceeds Many Solid Tumors	GLOBAL TAM \$60B+ Market Heart Failure Therapeutics
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Asset Program	Target Biology / Modality	Target Indication	Commercial Strategy	Current Development Status
NL-101 (Lead)	SIRT1+SIRT6+GATA4+ZBTB16 (mRNA-LNP)	Post-MI Ischemic Heart Failure (NYHA II-IV)	Wholly Owned / Clinical Lead	In Silico Complete → iPSC-CM Ready
NL-102	VE-Cadherin / CD31 Epigenetic Modulator	Ischemic Microvascular Angina	Internal Pipeline Expansion	In Silico Target Discovery
NL-103	Autonomic Synaptic Preservation Cocktails	Age-Related Cardiac Denervation / Arrhythmia	Biopharma Co-Development	In Silico Target Discovery

TWO-PRONGED BIOPHARMA VALUE CREATION MODEL:

- **Proprietary Asset Development (NL-101):** Retain 100% worldwide commercial rights through Phase 1/2a clinical proof-of-concept, positioning Nilus Lab for an M&A; exit with major cardiovascular pharma (Novartis, AstraZeneca, Pfizer).
- **Zenith Discovery Partnerships:** License platform target-discovery capabilities to biopharma partners exploring non-cardiac tissue targets, generating non-dilutive upfront licensing fees and research milestone payments.

PRE-SEED FINANCING: \$500,000 USD SAFE

Directly Funding the Exact In Vitro De-Risking Milestones Demanded by Institutional Investors

ROUND STRUCTURE: \$500,000 PRE-SEED SAFE

Instrument: Y Combinator Standard Post-Money SAFE • Valuation Cap: \$5,000,000 USD • Target Close: Q4 2026

Lead Investor Focus: Longevity & Biotech Angels, Family Offices, and Specialized Deep-Tech Seed Syndicates.

1. WET-LAB CRO ASSAYS \$250,000 (50% of Raise) • 5-arm human iPSC-CM combinatorial drop-out study. • Axion Maestro MEA field potential & conduction velocity. • TIME-seq targeted bisulfite sequencing for DNAm age reset.	2. mRNA / LNP SYNTHESIS \$125,000 (25% of Raise) • Modified mRNA synthesis (m1Ψ + degron tags). • VCAM-1 / NCAM-1 surface-conjugated LNP formulation. • Colloid stability, PDI, and encapsulation QC.
3. INTELLECTUAL PROPERTY \$75,000 (15% of Raise) • Global PCT patent filing on NL-101 Composition of Matter. • Method-of-use claims for non-arrhythmogenic reprogramming. • Freedom-to-operate (FTO) formalization.	4. PLATFORM COMPUTE & OPS \$50,000 (10% of Raise) • Cloud GPU cluster for single-cell scVI manifold updates. • Closed-loop data moat ingestion engine. • Operational runway through Month 18.

18-MONTH VALUE ACCELERATION TIMELINE:

- **Month 3:** mRNA synthesis and targeted LNP formulation complete.
- **Month 6:** Axion MEA electrophysiology readouts delivered; zero arrhythmia confirmed in human iPSC-CMs.
- **Month 9:** 5-arm factor drop-out assay completed; necessity of all 4 factors validated.
- **Month 12:** PCT patent published; comprehensive investor diligence data package locked.
- **Month 15–18:** Launch \$5M–\$10M Series Seed with leading biotech venture funds (Curie.Bio, ARCH, Khosla).

LEADERSHIP & SCIENTIFIC COLLABORATION NETWORK

Deep-Tech Computational Architecture Grounded in Stringent Translational Diligence

ALAA ALDEEN — FOUNDER & CEO

- Systems Biologist & Generative AI Architect (HomeLab Accelerator Cohort 5).
- Designed and trained Zenith v31.1 across 2.42M human cardiac single cells in 5,009-dimensional latent manifold.
- Built NEUROS-X 512-node spiking electrophysiological syncytium and automated Opentrons OT-2 / Boltz-1 pipelines.
- Driving translational strategy, investor engagement, and pre-seed execution.

INSTITUTIONAL SCIENTIFIC VALIDATION NETWORK

- **Accelerator Foundation:** HomeLab Accelerator Cohort 5 (Global scientific translation program).
- **Diligence Integration:** Rigorously audited against top-tier venture criteria (incorporating Curie.Bio evaluation on sarcomeric integrity and multi-factor drop-out necessity).
- **Benchmark Foundation:** Grounded in Human Cardiac Cell Atlas benchmarks (Litvišuková et al. *Nature* 2020) and EnsembleAge clocks (Haghani et al. *GeroScience* 2026).
- **Advisory Synergies:** Actively engaging academic leaders across cardiac electrophysiology and longevity medicine.

CORE GOVERNANCE & EXECUTION DISCIPLINES:

1. **Computational Manifold Dynamics:** High-dimensional latent space perturbation, continuous single-cell manifold learning.
2. **Cardiac Electrophysiology:** Optical mapping, field potential recordings, Connexin-43 gap junction preservation.
3. **Nucleic Acid Drug Delivery:** Active-targeted mRNA-LNP formulation, tissue tropism engineering, degron kinetics.
4. **Translational Cardiovascular Medicine:** Post-MI heart failure clinical trial design and regulatory surrogate endpoints.

INVESTING IN THE FIRST SAFE REJUVENATION THERAPEUTIC

We Don't Just Reverse the Clock. We Protect the Heart.

WHY NILUS LAB WILL WIN:

1. **The First Safe Target:** While competitors pursue unconstrained partial reprogramming and risk catastrophic arrhythmia, Nilus Lab is hyper-specialized on the heart with built-in electrophysiological safety gates.
2. **Scientifically De-Risked Lead:** NL-101 was nominated through a rigorous 10,000-candidate in silico filter, proven across a 512-node electrical syncytium, and backed by a clear 5-arm combinatorial drop-out plan.
3. **Immediate Catalyst:** Your \$500,000 Pre-Seed investment funds the definitive human iPSC-CM wet-lab assay, answering top-tier venture diligence criteria and unlocking a \$10M institutional Series Seed.

FOUNDER & CEO Alaa Aldeen info@niluslab.com	WEB PLATFORM niluslab.com Clinical Console & API	ROUND TERMS \$500,000 USD YC SAFE (\$5.0M Cap)	INVESTOR PORTAL Active Diligence Data Room Available
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